

# TB-500

Thymosin Beta-4 Fragment

|                     |              |             |           |          |
|---------------------|--------------|-------------|-----------|----------|
| SYSTEMIC<br>HEALING | INFLAMMATION | FLEXIBILITY | ENDURANCE | RECOVERY |
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## WHAT IS IT?

TB-500 is a synthetic fragment of Thymosin Beta-4 (Tβ4), a naturally occurring 44-amino-acid peptide found in virtually every cell in the body. It promotes systemic healing, reduces inflammation, and improves mobility — particularly in connective tissue.

## DOSING AT A GLANCE

| Loading (2x/wk) | Maintenance (1x/wk) | Loading Phase | Ongoing Maintenance |
|-----------------|---------------------|---------------|---------------------|
| 2–2.5 mg        | 1–1.5 mg            | 4–6 weeks     | Monthly             |

## KEY BENEFITS

- Systemic healing — works on muscles, tendons, ligaments, skin, eyes
- Significant reduction in inflammation and pain
- Improves flexibility and range of motion in injured tissue
- Promotes cardiomyocyte migration and cardiac repair after injury
- Accelerates wound healing and promotes skin repair
- Stimulates stem cell differentiation and migration
- Reported improvements in hair growth and re-pigmentation
- Neuroprotective effects — recovery from stroke or TBI
- Reduces adhesions and scar tissue formation

## HOW IT WORKS

TB-500 works primarily by binding to actin — a protein critical for cell structure and movement. This binding promotes cell migration and proliferation in injured tissue. It upregulates actin genes in healing cells and modulates inflammatory cytokines, reducing TNF-alpha and IL-6 while promoting anti-inflammatory pathways.

## PROTOCOLS

- **Loading Phase:** 2–2.5 mg SC/IM 2x per week for 4–6 weeks.

- **Maintenance:** 1–1.5 mg once weekly ongoing, or monthly once healed.
- **Acute Injury:** 2.5 mg 2x/week for 2–3 weeks, then reassess.
- **Stack:** TB-500 + BPC-157 = the gold standard injury healing stack.

## SIDE EFFECTS

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- Generally very well tolerated at therapeutic doses
- Fatigue or lethargy (common early, usually transient)
- Injection site reactions (mild)
- Theoretical concern: may promote cancer cell migration (not shown clinically at peptide doses)
- Headache (rare)

## STORAGE

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Lyophilized: room temp/refrigerator. Reconstituted: refrigerate, use within 30 days.

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■ **Not FDA approved. Banned by WADA for competitive athletes. Research use only.**

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